NETWORK TESTING TOOLS THEORY

1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Steps:-

- Open Command Prompt by pressing Win + R, typing cmd, and pressing Enter.
- Type the following command:

ping www.youtube.com

- Press Enter and wait for the results.
- At the bottom, check the Average value under the ping statistics. This is the average response time.

```
Pinging www.youtube.com [2001:4860:482b:400::] with 32 bytes of data:
Reply from 2001:4860:482b:400::: time=320ms
Reply from 2001:4860:482b:400::: time=125ms
Reply from 2001:4860:482b:400::: time=131ms
Reply from 2001:4860:482b:400::: time=133ms
```

2. Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes through to reach the destination.

Steps:

- Open Command Prompt (Win + R → type cmd → press Enter).
- Type this command:

  tracert www.flipkart.com

- Press Enter and wait for the trace to complete.
- Record each numbered hop shown in the output, including its IP address or hostname.

```
1    82 ms   203 ms   100 ms  2409:40c1:2145:9eb5::8a
2     *        *        *     Request timed out.
3   335 ms    98 ms   100 ms  2405:200:5210:4:3925::1
4    38 ms   114 ms    92 ms  2405:200:824:3632:62::6
5   198 ms   202 ms   100 ms  64:ff9b::c0a8:e3c2
6   111 ms    99 ms   101 ms  64:ff9b::c0a8:a872
```

3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

NSLookup: www.zomato.com

Steps :-

1. Open Command Prompt.
2. Type nslookup www.zomato.com and press Enter.
3. Record the IP address returned by your DNS server.

IP Address:-23.212.0.103

4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Ping Test to a Local Network Device

Steps:

- Connect your computer and your friend's laptop or phone to the same Wi-Fi network.
- Find the other device's IP address in its network settings.
- Open Command Prompt on your computer.
- Type the command (replace the IP with the device's actual IP):

  ping 192.168.1.5

- Press Enter and check the result.

5.Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.

Network Troubleshooting Report

- Ping: Useful for checking connectivity. In my test, I used ping www.youtube.com to check whether the website responded and to measure response time.
- Traceroute: Useful for diagnosing a website that is not loading. I used tracert www.flipkart.com to identify the network hops along the route.
- Nslookup: Useful for investigating DNS issues. I used nslookup www.zomato.com to find the domain's IP address.

- Website not loading: Ping and traceroute help check connectivity and identify possible network problems.
- Slow connection: Ping helps measure response time, while traceroute can help locate delays along the route.
- Suspected DNS issues: Nslookup checks whether the domain name resolves to an IP address.